

A Peep Into The Lesser-Known World Of FPGAs And ASICs

FPGAs and ASICs are widely used nowadays, especially in electronics equipment such as laptops, smartphones, and television sets. The components have their strengths and weaknesses. Their suitability depends on the type of application. With time, it is hoped that FPGAs and ASICs will undergo new advancements and efficiently meet customer needs

Abhinav Behl

FPGAs and ASICs form the core of modern-day electronics systems. Without them much of the equipment that we take for granted would cease to exist. But there is more to what meets the eye. Behind the state-of-the-art processors, months and years of effort makes them robust and reliable. So let us find out more about their design journey.

Application specific integrated circuit, or ASIC, is a type of IC that is used only for specific applications. Field programmable gate array, or FPGA, is an IC that can be repeatedly used for different applications as it is reprogrammable. To do so, one needs to rewrite the Hardware Design Language (HDL) code and reprogram the FPGA accordingly.

Some other differences between them are:

Time to market. ASIC is notoriously slow when it comes to time to market. A good ASIC may take five to six years to get fully developed from scratch. On the other hand, FPGA projects such as PCI and USB require eight months to a year to get ready.

Unit cost. The per-unit cost of FPGA is high as compared to ASIC. For that reason, FPGAs get developed only when a large number needs to be produced.

Power consumption. Nowadays, power consumption is a big issue. We want our elec-

tronic devices to save battery power and work efficiently, even on low power. From this perspective, FPGA is not a good choice since it has high power consumption rate compared to ASIC.

Frequency. FPGA operates at a much less frequency as compared to an ASIC. The operational difficulty increases at higher frequencies. For an ASIC, however, there is no specific frequency speed.

Design complexity. An ASIC is complicated to design and manufacture, therefore it takes a long time to market. But FPGA is comparatively simpler.

FPGA design flow

The process of designing an FPGA generally begins with functional specifications at the top. Next, the HDL code is written on which behavioural simulation is performed. After that come the steps of synthesis, place & route, download and verify, along with static timing analysis, which are handled by the designing software.

For a Xilinx FPGA, the Vivado software is used. For an Intel FPGA, Quartus is used. In Vivado, the step Place & Route is also known as Implementation.

During behavioural simulation, the HDL code is checked for its accuracy and expected behaviour, that is, whether the code has any functional bug in it or not.

During static timing analysis, the setup time and hold time violations are checked. Although hold time delay can be a bit difficult to resolve, setup time is easy if you

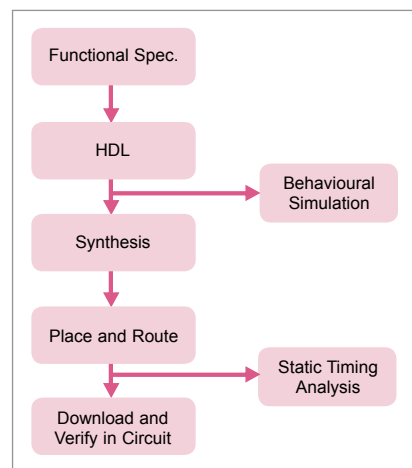


Fig. 1: Design flow of FPGA